Stroma World — Biological Modules v1.0

> **What this document defines:** The minimal set of functional units that, wired together, produce the stroma barrier and all behaviors the Reality Anchors require. **Design rule:** Each module does exactly one thing. If a module has two purposes, split it. If two modules always fire together, merge them. **Implementation target:** Each module maps to one function or one clearly bounded block in PhysiCell's `custom.cpp` + XML config.

Module Architecture Overview

```
MICROENVIRONMENT (PDE fields)
┌──────────────────────────────────────────────────────────────┐
│ O₂   TGF-β   SHH   Drug   ECM_density                        │
└──────────────────────────────────────────────────────────────┘
      │        │        │        │
  ┌───┴────┐ ┌─┴──────┐ ┌─┴──────┐ ┌─┴──────────┐
  │ Module 1 │ │ M2   │ │ M3   │ │ Module 7   │
  │ O₂       │ │Para- │ │Stroma│ │ Drug       │
  │ Sensing  │ │crine │ │Activ.│ │ Response   │
  └────────┬─┘ └──────┘ └──────┘ └────────────┘
      │        │        │        │
  ┌──────────────────────────────────────────────────────────┐
  │                CELL DECISION LAYER                       │
  │          (Boolean network + combination rules)           │
  └──────────────────────────────────────────────────────────┘
      │        │        │        │
  ┌───┴────┐ ┌─┴──────┐ ┌─┴──────┐ ┌─┴──────────┐
  │ Module 4 │ │ M5   │ │ M6   │ │ Module 8   │
  │ Prolif & │ │ EMT  │ │ ECM  │ │ Mechanical │
```

```
Death    Engine   Prod.   Compaction
```

Eight modules. Three read the environment. Four write to the environment. One (the cell decision layer) is the internal routing that connects readers to writers. Every arrow in the Reality Anchor causal chain passes through at least one of these modules.

Module 1: Oxygen Sensing & Hypoxia Response

Purpose

Translate local oxygen concentration into an intracellular hypoxia state that modifies downstream behaviors. This module creates the hypoxia gradient that drives EMT, stress adaptation, and the stroma-amplifying positive feedback loop.

Inputs

Input	Source	Type
Local O_2 concentration	Microenvironment O_2 field, sampled at cell position	Continuous (from PDE)
hypoxia_response_threshold	Knob 6a	Fixed parameter

Internal Logic

```
IF local_O2 < threshold mapped from Knob 6a:
  HIF1A = ON
ELSE:
  HIF1A = OFF
```

Binary switch. No intermediate states in v1.0.

Outputs

Output	Destination	Effect
HIF1A state (ON/OFF)	Cell decision layer	Feeds into EMT induction (Module 5), stress adaptation, secretion amplification
Phenotype shift magnitude	Determined by Knob 6b	LOW/MED/HIGH controls how much downstream behaviors change when HIF1A=ON

What It Changes in the Environment

Nothing directly. This is a sensing module. It reads O_2 but doesn't modify any field. Its effects on the environment are indirect, via downstream modules that change secretion rates (Module 2) or ECM production (Module 6).

Reality Anchors Served

Anchor 5 (hypoxia develops centrally), Anchor 6 (EMT is spatial — hypoxia gradient drives the spatial restriction of EMT).

Module 2: Paracrine Secretion Engine

Purpose

Govern the rate at which tumor cells release TGF-β and SHH into the microenvironment. This is the root cause of barrier formation — the module that makes the tumor build its own prison.

Inputs

Input	Source	Type
KRAS state	Gene state (always GOF in canonical PDAC)	Fixed
HIF1A state	Module 1 output	ON/OFF
tgfb_secretion_rate	Knob 1a	0–1
shh_secretion_rate	Knob 1b	0–1
Cell type	Cell identity flag	Tumor vs. Stromal

Internal Logic

```
BASE secretion rates set by Knobs 1a, 1b.

IF cell is TUMOR:
   TGF-β secretion = base rate from Knob 1a
   SHH secretion   = base rate from Knob 1b
   IF HIF1A = ON:
      TGF-β secretion → amplified (shift up one qualitative tier)
      SHH secretion   → unchanged (SHH is not hypoxia-responsive)

IF cell is STROMAL and ACTA2 = ON:
   TGF-β secretion = moderate (CAFs produce TGF-β, creating positive feedback)
   SHH secretion   = 0 (stromal cells do not secrete SHH)

IF cell is STROMAL and ACTA2 = OFF:
   All secretion = 0 (quiescent PSCs are silent)
```

Outputs

Output	Destination	Effect
TGF-β flux	Microenvironment TGF-β field	Adds TGF-β at cell position; diffuses outward via PDE
SHH flux	Microenvironment SHH field	Adds SHH at cell position; diffuses outward via PDE

What It Changes in the Environment

TGF-β field — increased concentration radiating from tumor cells (and activated CAFs). Creates the gradient that activates stroma. **SHH field** — increased concentration radiating from tumor cells only. Second arm of stromal activation.

Reality Anchors Served

Anchor 1 (barrier self-assembles — this module provides the signal), Anchor 3 (SHH inhibition — setting Knob 1b to 0 tests the SHH arm), Anchor 4 (SMAD4 asymmetry — TGF-β must be present for the asymmetry to manifest).

Module 3: Stromal Activation Switch

Purpose

Determine when and whether a quiescent pancreatic stellate cell (PSC) transitions to an activated cancer-associated fibroblast (CAF). This is the receiver side of the paracrine loop — Module 2 sends the signal, Module 3 decides if the stroma hears it.

Inputs

Input	Source	Type
Local TGF-β concentration	Microenvironment TGF-β field, sampled at cell position	Continuous
Local SHH concentration	Microenvironment SHH field, sampled at cell position	Continuous
Current ACTA2 state	Cell internal state	ON/OFF
Activation threshold	Stromal calibration parameter (future Stromal Knobs doc)	Fixed

Internal Logic

```
IF cell is STROMAL and ACTA2 = OFF (quiescent PSC):
  combined_signal = f(local_TGF-β, local_SHH)
  — f is additive: either signal alone can activate, both together activate faster/more reliably
  IF combined_signal > activation_threshold:
    ACTA2 → ON (irreversible in v1.0)
    GLI1 → ON if SHH component contributed
    Cell identity transitions: PSC → CAF

IF cell is STROMAL and ACTA2 = ON (already activated CAF):
  No state change. Activation is permanent in v1.0.
  GLI1 tracks ongoing SHH exposure (reversible — drops if SHH disappears).
```

v1.0 simplification: CAF activation is a one-way switch. Once activated, a CAF does not revert to quiescence. This is biologically approximate (some CAF plasticity exists) but avoids oscillatory behavior that would complicate early debugging. Reversion can be added in v1.1 if needed.

Outputs

Output	Destination	Effect
ACTA2 state (ON/OFF)	Cell decision layer, Module 6 (ECM production)	Determines whether this cell produces ECM
GLI1 state (ON/OFF)	Cell decision layer	Modulates CAF proliferation rate and ECM output
CAF identity flag	Cell type register	Marks cell as activated for all downstream modules

What It Changes in the Environment

Nothing directly. Like Module 1, this is a sensing/decision module. Its environmental effects come through Module 6 (ECM production) and Module 2 (CAF TGF-β secretion feedback).

Reality Anchors Served

Anchor 1 (self-assembly — PSCs must activate), Anchor 3 (SHH inhibition — GLI1 arm is removed but TGF-β arm remains), Anchor 7 (ECM rebuilds after degradation — CAFs persist and re-activate production).

Module 4: Proliferation & Death Arbitration

Purpose

Resolve the net growth behavior of each cell at each time step: does it divide, die, or hold? This module integrates growth drive, checkpoint status, death resistance, and drug effects into a single proliferation/death decision.

Inputs

Input	Source	Type
Gene states	KRAS, MYC, CDKN2A, TP53, BCL-XL, BAX, SMAD4	Fixed per cell
HIF1A state	Module 1	ON/OFF
ZEB1 state	Module 5	ON/OFF (go-vs-grow tradeoff)
Local drug concentration	Microenvironment drug field	Continuous
ABCB1 state	Module 7	ON/OFF
Local O_2	Microenvironment O_2 field	Continuous
Mechanical pressure	Module 8 / PhysiCell mechanics	Continuous
Cell type	Cell identity	Tumor vs. Stromal

Internal Logic

```
STEP 1: Determine growth drive qualitative tier
  — Apply combination rules from Gene State doc (KRAS, MYC, EGFR contributions)
  — Apply go-vs-grow penalty if ZEB1=ON (mild proliferation reduction)
  — Apply hypoxia modifier if HIF1A=ON (acute: slow; chronic: adapted)
  → Output: VERY HIGH / HIGH / MODERATE / BASELINE / LOW

STEP 2: Determine checkpoint status
  — Apply veto rules (CDKN2A=LOF + TP53=LOF + MYC=GOF → NONE)
```

```
  — Check SMAD4 status for TGF-β brake (Module cross-reference)
  → Output: NONE / LOW / MODERATE / HIGH

STEP 3: Determine death resistance
  — Apply combination rules (TP53, BCL-XL, BAX)
  — Add drug-conditional ABCB1 contribution (Module 7)
  — If drug present AND ABCB1=ON: effective drug concentration reduced
  → Output: VERY HIGH / HIGH / MODERATE / LOW

STEP 4: Resolve
  — Proliferation rate = map(growth_drive, checkpoint) → PhysiCell cycle.transition_rate
  — Apoptosis rate = map(death_resistance, effective_drug_concentration) → PhysiCell death.apoptosis_rate
  — If mechanical pressure exceeds threshold: proliferation paused (contact inhibition)

FOR STROMAL CELLS:
  — CAFs (ACTA2=ON): moderate proliferation, boosted if GLI1=ON
  — PSCs (ACTA2=OFF): quiescent, minimal proliferation
  — Stromal cells do not respond to the drug substrate in v1.0
```

Outputs

Output	Destination	Effect
cycle.transition_rate	PhysiCell cell cycle	Controls division timing
death.apoptosis_rate	PhysiCell death model	Controls probability of death per time step
Contact inhibition flag	PhysiCell mechanics	Pauses cycle under high pressure

What It Changes in the Environment

Indirectly, everything. A dividing cell creates a new cell (which secretes, consumes O_2, exerts pressure). A dying cell is removed (reducing secretion, opening space). But this module doesn't directly write to any

microenvironment field — it modifies PhysiCell's intrinsic cell cycling and death parameters.

Reality Anchors Served

Anchor 3 (SHH inhibition — tumor grows faster when unconfined), Anchor 4 (SMAD4 asymmetry — growth arrest arm), Anchor 9 (barrier predicts drug failure — death arbitration must be drug-concentration-dependent), Anchor 10 (sanctuary — survivors must retain proliferative capacity for regrowth).

Module 5: EMT Engine

Purpose

Manage the epithelial-to-mesenchymal transition for tumor cells. Determines when a cell goes mesenchymal, what changes, and ensures spatial restriction (periphery, not core).

Inputs

Input	Source	Type
Local TGF-β concentration	Microenvironment TGF-β field	Continuous
SMAD4 state	Gene state	WT/LOF
HIF1A state	Module 1	ON/OFF
emt_induction_threshold	Knob 5a	0–1
emt_phenotype_extent	Knob 5b	L/M/H

Internal Logic

```
COMPUTE induction_signal:
  — TGF-β component: local TGF-β concentration (SMAD-independent arm; works regardless of SMAD4)
  — Hypoxia component: if HIF1A=ON, add boost to induction signal
  — Combined: additive. Either can trigger EMT alone; both together make it easier.

IF induction_signal > threshold from Knob 5a:
  ZEB1 → ON
  CDH1 → OFF
  Apply phenotype changes per Knob 5b extent:
    LOW:    motility ↑ slight, adhesion ↓ slight, MMP2 off
    MEDIUM: motility ↑ moderate, adhesion ↓ moderate, MMP2 on
    HIGH:   motility ↑ maximum, adhesion ↓ minimum, MMP2 on, PD-L1 tracked
ELSE:
  ZEB1 → OFF
  CDH1 → ON
  Epithelial phenotype: baseline motility, normal adhesion
```

v1.0 simplification: EMT is reversible — if the cell moves to a region with low TGF-β and adequate O_2, it can revert to epithelial. This allows the spatial restriction pattern (Anchor 6) to emerge dynamically rather than requiring permanent commitment.

Outputs

Output	Destination	Effect
ZEB1 state	Module 4 (go-vs-grow), Module 2 (no direct effect)	Modifies proliferation via tradeoff
motility.speed	PhysiCell motility	Controls migration rate

Output	Destination	Effect
cell_cell_adhesion_strength	PhysiCell mechanics	Controls how strongly cells stick together
MMP2 activity flag	Module 6 (ECM degradation arm)	Enables local ECM digestion
IET contribution	Bookkeeping (Cycle 2 prep)	PD-L1 status tracked but no immune cells to interact with

What It Changes in the Environment

ECM density field (indirect, via MMP2). When MMP2 is active, the cell degrades ECM in its immediate neighborhood. This is the *counter-signal* to Module 6's ECM production — the tumor can locally thin the barrier at its invasive front.

Reality Anchors Served

Anchor 6 (EMT is peripheral — spatial signal gradients drive peripheral EMT), Anchor 7 (ECM degradation — MMP2 from EMT cells is one source of local barrier thinning).

Module 6: ECM Production & Remodeling

Purpose

Control the creation and destruction of extracellular matrix. This module builds the physical barrier. It is the only module that directly increases ECM density.

Inputs

Input	Source	Type
ACTA2 state	Module 3	ON/OFF (only CAFs produce ECM)

Input	Source	Type
GLI1 state	Module 3	ON/OFF (boosts production if SHH pathway active)
MMP2 activity	Module 5	ON/OFF (degrades local ECM)
External degradation event	EA intervention (simulating hyaluronidase, collagenase)	Event trigger

Internal Logic

```
PRODUCTION (stromal cells only):
  IF ACTA2 = ON:
    HA_production_rate  = HIGH (HAS2 active)
    COL_production_rate = HIGH (COL1A1 active)
    IF GLI1 = ON:
      Both rates → boosted (SHH arm adds to TGF-β-driven production)
  IF ACTA2 = OFF:
    All production = 0

DEGRADATION:
  IF MMP2 active at this location (from any cell):
    Local ECM_density -= degradation amount per time step
  IF external degradation event active:
    Global or regional ECM_density -= intervention amount

NET CHANGE per time step per location:
  ΔECM = Σ(production from nearby CAFs) - Σ(degradation from nearby MMP2+ cells) - external degradation

ECM_density is floored at 0 (cannot go negative).
```

Implementation note on two-component ECM: Internally, HA and collagen are tracked as contributions to a single `ECM_density` field but with *different effects on physics*:

- HA contribution → modifies drug diffusion coefficient (interstitial pressure / gel resistance)
- Collagen contribution → modifies mechanical stiffness (structural confinement) Both are produced by the same CAFs but can be independently targeted by the EA (HAS2 inhibition vs. collagenase). For v1.0, this can be implemented as two scalar attributes on each ECM field voxel rather than two separate PDE fields — avoids doubling PDE computation cost.

Outputs

Output	Destination	Effect
ECM_density field update	Microenvironment	Modifies drug diffusion, O_2 diffusion, and mechanical resistance at every voxel
HA component fraction	Physics calculations	Weights the diffusion-impeding effect
Collagen component fraction	Physics calculations	Weights the mechanical confinement effect

What It Changes in the Environment

ECM_density field — the primary barrier variable. This is the only module that directly writes to the ECM field. Everything else that affects the barrier does so by modifying inputs to this module (activating CAFs, triggering MMP2, applying degradation interventions).

Additionally, changes in ECM_density cascade to:

- Drug diffusion coefficient (higher ECM → slower drug diffusion)
- O_2 diffusion coefficient (higher ECM → slower O_2 diffusion → more hypoxia)

- Mechanical stiffness seen by Module 8

Reality Anchors Served

Anchor 1 (barrier self-assembles — this module builds it), Anchor 2 (drug diffusion drops — ECM impedes drug), Anchor 5 (hypoxia — ECM impedes O_2), Anchor 7 (ECM degradation and rebuilding), Anchor 8 (HA vs. collagen distinction), Anchor 9 (barrier density predicts drug failure), Anchor 10 (sanctuary — dense ECM protects interior cells).

This is the most-referenced module in the anchor suite. If Module 6 is broken, most anchors fail.

Module 7: Drug Response & Adaptive Resistance

Purpose

Handle drug pharmacokinetics at the single-cell level: uptake, intracellular accumulation, efflux pump induction, and the resulting effect on cell viability. Separates "did the drug get here" (a microenvironment question) from "did the drug kill the cell" (this module's question).

Inputs

Input	Source	Type
Local drug concentration	Microenvironment drug field, sampled at cell position	Continuous
efflux_induction_delay	Knob 7a	0–1
efflux_strength	Knob 7b	0–1
NRF2 state	Activated by drug-induced stress or HIF1A	ON/OFF
ABCB1 state	Downstream of NRF2	ON/OFF
Time since first drug exposure	Cell internal clock	Continuous

Input	Source	Type
apoptosis_resistance	Knob 4 (baseline)	0–1

Internal Logic

```
STEP 1: Drug uptake
  — Cell absorbs drug proportional to local extracellular concentration
  — Intracellular drug accumulates over time

STEP 2: Stress sensing
  — IF intracellular drug > stress threshold:
    NRF2 → ON (priming begins)
  — Start induction timer

STEP 3: Efflux activation
  — IF NRF2 = ON AND time_since_exposure > delay mapped from Knob 7a:
    ABCB1 → ON
  — IF ABCB1 = ON:
    Intracellular drug reduced by factor mapped from Knob 7b per time step
    (drug is pumped out — effective concentration drops)

STEP 4: Death signal
  — Effective drug kill signal = f(intracellular drug concentration, apoptosis_resistance)
  — Pass effective kill signal to Module 4 (Proliferation & Death Arbitration)
  — Module 4 integrates this with baseline death resistance to set apoptosis rate

WHEN DRUG IS ABSENT:
  — All of the above is inert. ABCB1 provides zero benefit.
  — NRF2 may deactivate over time if stress signal is removed (slow decay, not instant).
```

Outputs

Output	Destination	Effect
ABCB1 state	Module 4 (death resistance contribution)	Reduces effective drug-induced death
Intracellular drug concentration	Module 4 (kill signal)	Drives apoptosis when high enough
Drug consumption	Microenvironment drug field	Cell uptake depletes local extracellular drug (small sink effect)

What It Changes in the Environment

Drug field — mild depletion. Cells that take up drug act as sinks, slightly reducing local extracellular concentration. This is a minor effect compared to ECM-mediated diffusion blocking, but matters in densely packed regions where many cells compete for limited drug.

Reality Anchors Served

Anchor 7 (ECM degradation + drug — drug must actually kill when it arrives), Anchor 9 (drug failure — resistance machinery contributes to treatment failure beyond just diffusion blocking), Anchor 10 (sanctuary population — cells in low-drug zone never activate ABCB1 and remain drug-naive for when drug is re-applied; cells that did experience drug have activated efflux).

Module 8: Mechanical Compaction

Purpose

Translate tumor cell proliferation and ECM stiffness into mechanical pressure that feeds back on cell behavior and barrier density. This is the physical (non-biochemical) arm of barrier formation.

Inputs

Input	Source	Type
Local cell density	PhysiCell mechanics engine (cell neighbor count / volume fraction)	Continuous
ECM stiffness	Module 6 output (collagen component)	Continuous
mechanical_compaction_strength	Knob 8	0–1
Cell proliferation state	Module 4	Dividing / quiescent / arrested

Internal Logic

```
STEP 1: Compute local solid stress
  — Stress = f(cell_density, ECM_stiffness, Knob 8 scaling)
  — Higher cell density in stiff ECM → higher stress
  — This leverages PhysiCell's built-in mechanics for cell-cell pushing,
    extended with ECM resistance

STEP 2: Feedback to cells
  — IF solid stress > contact_inhibition_threshold:
      → Proliferation paused (fed to Module 4)
  — This primarily affects tumor cells at the tumor-stroma boundary
    where expanding tumor pushes against rigid ECM

STEP 3: Feedback to ECM
  — Proliferative pressure from growing tumor compresses adjacent ECM
  — Local ECM_density += compaction increment proportional to stress and Knob 8
  — This is PASSIVE densification — not CAF production, just physical compression
```

Outputs

Output	Destination	Effect
Solid stress value	Module 4 (contact inhibition)	Can pause proliferation under high compression
ECM compaction increment	Module 6 / ECM_density field	Passively increases barrier density at tumor-stroma boundary
Cell displacement forces	PhysiCell mechanics	Cells push each other; tumor expands against resistance

What It Changes in the Environment

ECM_density field — passive increase at tumor-stroma boundary. This is the mechanical complement to Module 6's biochemical ECM production. Both contribute to barrier density, but through independent mechanisms. Module 6 adds new ECM material (CAFs building matrix). Module 8 compresses existing ECM material (tumor growth squeezing the wall tighter).

O_2 and drug diffusion — indirect. Higher ECM density from compaction further impedes diffusion of all substrates, amplifying the effects already created by Module 6.

Reality Anchors Served

Anchor 3 (SHH paradox — mechanical confinement is part of why stroma restrains tumor; removing stroma releases this pressure), Anchor 8 (collagen component contributes to structural rigidity that this module leverages), Anchor 10 (solid stress contributes to vascular collapse at tumor margin, deepening the sanctuary).

Module Interaction Map

Which modules read from which:

Module 1 (O_2 Sensing)
  reads: O_2 field
  feeds: Module 2, Module 4, Module 5

Module 2 (Paracrine Secretion)
  reads: Module 1, Module 3 (CAF feedback), gene states
  feeds: TGF-β field, SHH field

Module 3 (Stromal Activation)
  reads: TGF-β field, SHH field
  feeds: Module 2 (CAF TGF-β feedback), Module 4, Module 6

Module 4 (Proliferation & Death)
  reads: Module 1, Module 3, Module 5, Module 7, Module 8, gene states
  feeds: PhysiCell cycle/death, Module 8 (proliferation drives compaction)

Module 5 (EMT Engine)
  reads: Module 1, TGF-β field, gene states
  feeds: Module 4 (go-vs-grow), Module 6 (MMP2 degradation)

Module 6 (ECM Production)
  reads: Module 3, Module 5, external interventions
  feeds: ECM_density field → O_2 diffusion, drug diffusion, Module 8

Module 7 (Drug Response)
  reads: Drug field, Module 1 (HIF1A → NRF2 priming)
  feeds: Module 4 (kill signal), drug field (sink)

Module 8 (Mechanical Compaction)
  reads: Module 4 (proliferation state), Module 6 (ECM stiffness)
  feeds: Module 4 (contact inhibition), ECM_density field

Feedback Loops Encoded in This Wiring

Loop	Path	Type	Anchor
Stroma-building loop	M2 → TGF-β → M3 → M6 → ECM ↑ → M3 (CAF TGF-β feedback) → M2	Positive	1, 9
Hypoxia amplification	M6 → ECM ↑ → O_2 diffusion ↓ → M1 → HIF1A → M2 (amplified secretion) → M6	Positive	5
Mechanical confinement	M4 (proliferation) → M8 → contact inhibition → M4 (arrest)	Negative	3
EMT invasion front	M1 → M5 → MMP2 → M6 (local ECM ↓) → O_2 locally ↑ → M1 (HIF1A off) → M5 (EMT reverts)	Negative (self-limiting)	6
Drug resistance induction	Drug → M7 → NRF2/ABCB1 → M4 (reduced kill) → cell survives → continues secreting (M2)	Positive (resistance)	10

Two positive, two negative, one adaptive. This is the minimal loop structure needed to produce all ten Reality Anchor behaviors.

Execution Order Per Time Step

Modules must execute in a specific order to avoid stale-data artifacts:

```
1. Module 1  — Sense O₂          (reads environment)
2. Module 3  — Sense TGF-β/SHH   (reads environment)
3. Module 7  — Sense drug        (reads environment)
   ——— all sensing complete ———
4. Module 5  — EMT decision      (reads Module 1 + environment)
5. Module 4  — Proliferation/Death (reads Modules 1,3,5,7,8)
   ——— all cell decisions complete ———
6. Module 2  — Secrete           (writes TGF-β, SHH fields)
7. Module 6  — Produce/degrade ECM (writes ECM field)
8. Module 8  — Mechanical update  (writes ECM field, reads Module 4)
   ——— all environment writes complete ———
9. PhysiCell PDE solver advances diffusion for all fields
10. PhysiCell mechanics solver resolves cell positions
```

Rule: All reads happen before all writes within each time step. No module reads a field that another module has already written to in the same step. This prevents order-dependent artifacts and ensures reproducibility.

What's NOT a Module (and Why)

Candidate	Why excluded
Immune response module	No immune cells in Cycle 1. IET is tracked as bookkeeping within Modules 4 and 5 but has no functional effect. Becomes Module 9 in Cycle 2.
Angiogenesis / vasculature module	No vessel dynamics in Cycle 1. O_2 and drug are supplied from the domain boundary as proxies. Becomes a module in Cycle 3 or when vasculature is added.
Metabolic reprogramming module	Warburg effect and nutrient competition are approximated by Module 1 (hypoxia response). Full metabolism is Cycle 3.

Candidate	Why excluded
Clonal evolution module	All tumor cells share one genotype in v1.0. Mutation and selection during the simulation is Cycle 3.
ECM alignment / fiber orientation	Collagen fiber alignment affects invasion directionality, but is a second-order spatial effect. ECM is isotropic in v1.0.

Each excluded candidate has a natural insertion point in the module architecture (the wiring diagram is designed to accommodate them) but adds complexity that isn't required to pass the 10 Reality Anchors.

Eight modules. Three read the environment. Four write to it. One routes decisions. They produce two positive feedback loops, two negative feedback loops, and one adaptive loop — the minimal wiring to reproduce all ten Reality Anchors.